THE MORPHISM FROM A VERSAL RING TO THE STACK IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 0DQU: sources/stacks-chow/tags.txt, source lines 17199–17199. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0DR1: sources/stacks-chow/tags.txt, source lines 17206–17206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DR0: sources/stacks-chow/tags.txt, source lines 17205–17205. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DQX: sources/stacks-chow/tags.txt, source lines 17202–17202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06PX: sources/stacks-chow/tags.txt, source lines 8586–8586. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06Q0: sources/stacks-chow/tags.txt, source lines 8589–8589. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DQY: sources/stacks-chow/tags.txt, source lines 17203–17203. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06EE: sources/stacks-chow/tags.txt, source lines 8213–8213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 01T9: sources/stacks-chow/tags.txt, source lines 2366–2366. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0DQU let $x_0 : \mathrm{Spec}(k) \to \mathcal{X}$ be a morphism, where k is a finite type field over S. Let A be a versal ring to $\mathcal{X}$ at x_0. Then the morphism $\mathrm{Spec}(A) \to \mathcal{X}$ of Remark 0DR1 is flat.*

Proof. If the local ring of S at the image point is a G-ring, then this follows immediately from Lemma 0DR0 and the fact that the map from a Noetherian local ring to its completion is flat. In general we prove it as follows.

Step I. If A and A' are two versal rings to $\mathcal{X}$ at x_0, then the result is true for A if and only if it is true for A'. Namely, after possible swapping A and A', we may assume there is a formally smooth map $\varphi : A \to A'$ such that the composition

$$\mathrm{Spec}(A') \to \mathrm{Spec}(A) \to \mathcal{X}$$

is the morphism $\mathrm{Spec}(A') \to \mathcal{X}$, see Lemma 0DQX and Remark 0DR1. Since $A \to A'$ is faithfully flat we obtain the equivalence from Morphisms of Stacks, Lemmas 06PX and 06Q0.

Step II. Let l/k be a finite extension of fields. Let $x_{l,0} : \mathrm{Spec}(l) \to \mathcal{X}$ be the induced morphism. Let A be a versal ring to $\mathcal{X}$ at x_0 and let $A \to A'$ be as in Lemma 0DQY. Then again the composition

$$\mathrm{Spec}(A') \to \mathrm{Spec}(A) \to \mathcal{X}$$

is the morphism $\mathrm{Spec}(A') \to \mathcal{X}$, see Remark 0DR1. Arguing as before and using step I to see choice of versal rings is irrelevant, we see that the lemma holds for x_0 if and only if it holds for $x_{l,0}$.

Step III. Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Then we can choose a finite type point z_0 on $Z = U \times_{\mathcal{X}} x_0$ (this is a nonempty algebraic space). Let $u_0 \in U$ be the image of z_0 in U. Choose a scheme and a surjective étale map $W \to Z$ such that z_0 is the image of a closed point $w_0 \in W$ (see Morphisms of Spaces, Section 06EE). Since $W \to \mathrm{Spec}(k)$ and $W \to U$ are of finite type, we see that $\kappa(w_0)/k$ and $\kappa(w_0)/\kappa(u_0)$ are finite extensions of fields (see Morphisms, Section 01T9). Applying Step II twice we may replace x_0 by $u_0 \to U \to \mathcal{X}$. Then we see our morphism is the composition

$$\mathrm{Spec}(\mathcal{O}^\wedge_{U,u_0}) \to U \to \mathcal{X}$$

The first arrow is flat because completion of Noetherian local rings are flat (Algebra, Lemma 00MB) and the second arrow is flat as a smooth morphism is flat. The composition is flat as composition preserves flatness. □